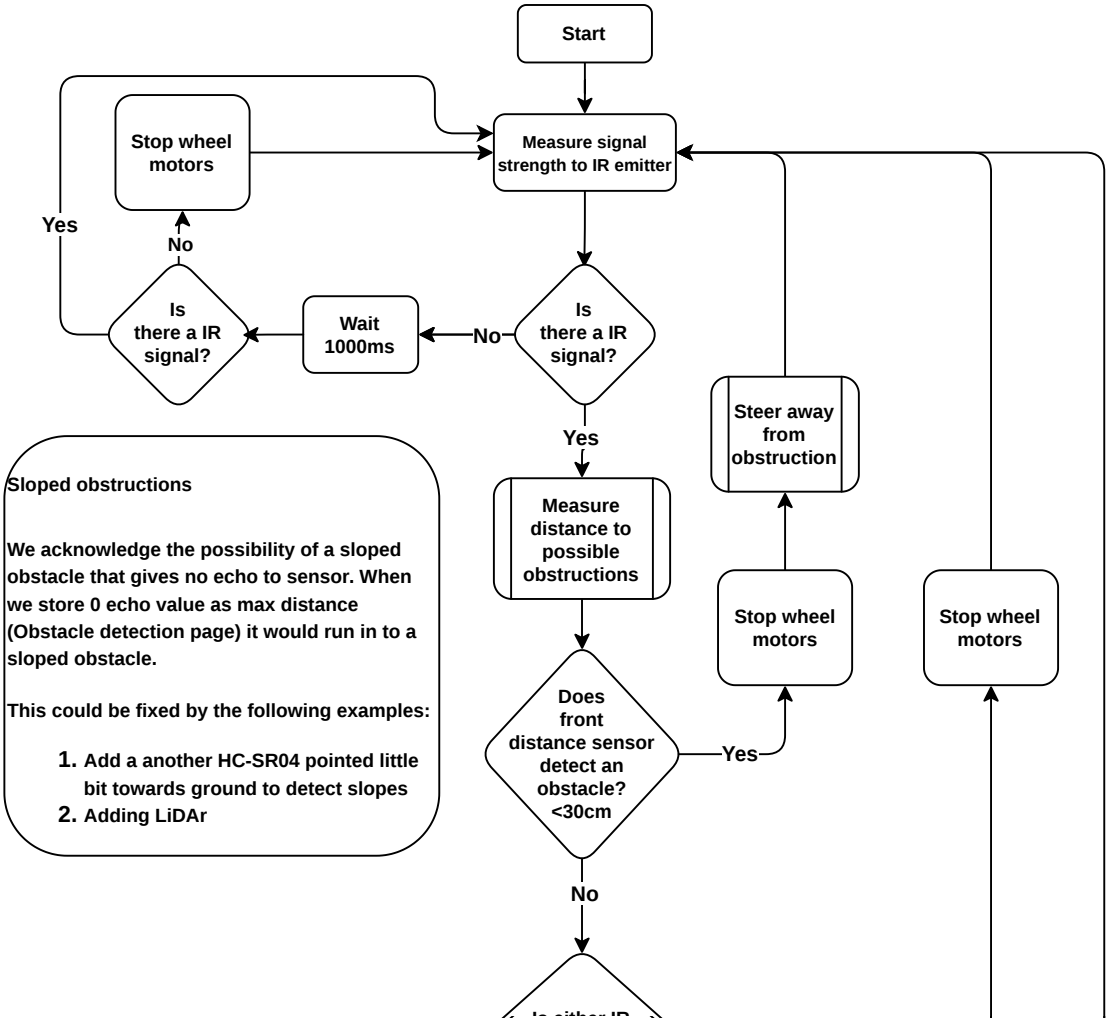
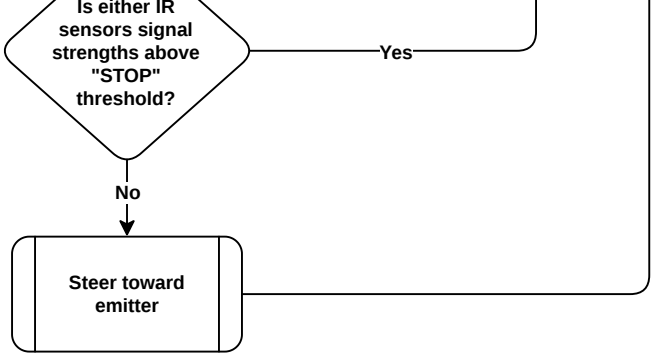
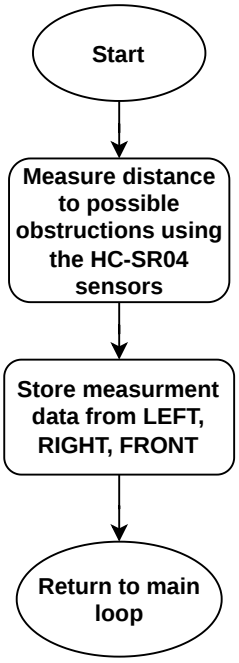
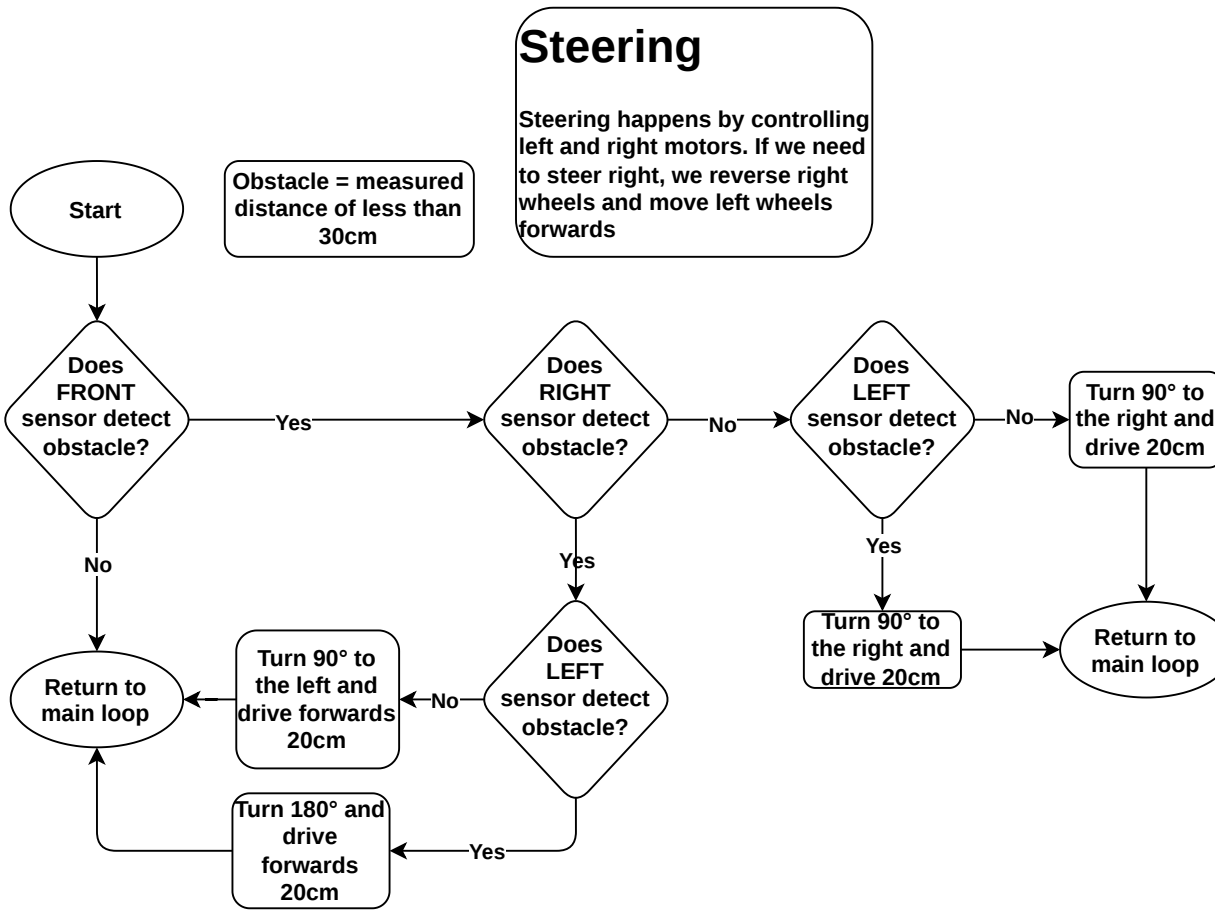

Start
Measure signal strength to IR emitter
Stop wheel motors
Yes
No
Is there a IR signal?
Wait 1000ms
No
Is there a IR signal?
Yes
Steer away from obstruction
Measure distance to possible obstructions
Stop wheel motors
Stop wheel motors
Does front distance sensor detect an obstacle? <30cm
Yes
No
Sloped obstructions
We acknowledge the possibility of a sloped obstacle that gives no echo to sensor. When we store 0 echo value as max distance (Obstacle detection page) it would run in to a sloped obstacle.
This could be fixed by the following examples:
1. Add a another HC-SR04 pointed little bit towards ground to detect slopes
2. Adding LiDAr

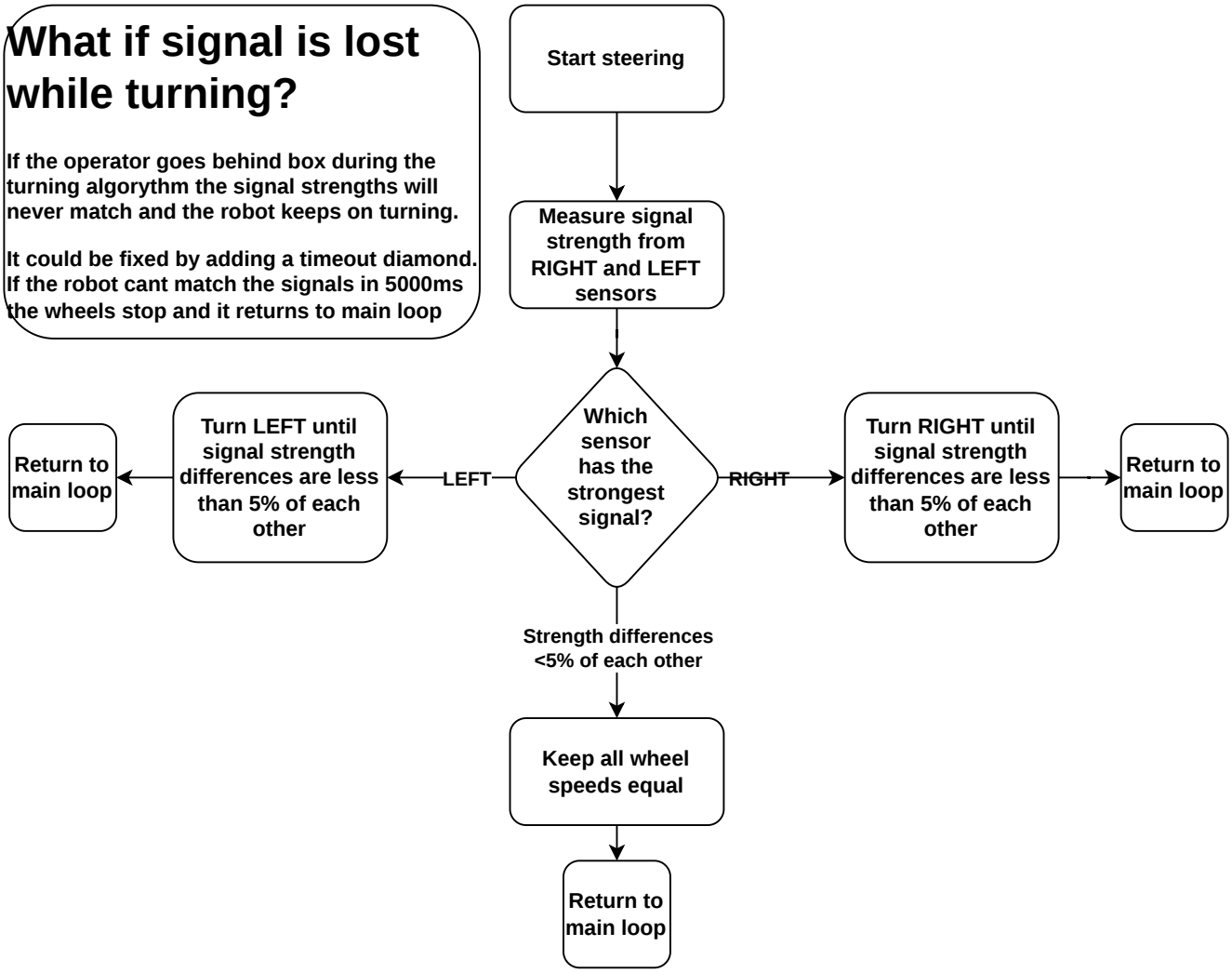

Is either IR sensors signal strengths above "STOP" threshold?
Yes
No
Steer toward emitter



What if no echo?

If a sensor has no echo (0), it will store it as max distance so the robot does not stop moving when it operates in a large empty area.


Steering
Steering happens by controlling left and right motors. If we need to steer right, we reverse right wheels and move left wheels forwards
Start
Obstacle = measured distance of less than 30cm
Does FRONT sensor detect obstacle?
Yes
Does RIGHT sensor detect obstacle?
No
Does LEFT sensor detect obstacle?
No
Turn 90° to the right and drive 20cm
No
Yes
Yes
Turn 90° to the right and drive 20cm
Return to main loop
Does LEFT sensor detect obstacle?
No
Turn 90° to the left and drive forwards 20cm
Return to main loop
Yes
Turn 180° and drive forwards 20cm


What if signal is lost while turning?
If the operator goes behind box during the turning algorythm the signal strengths will never match and the robot keeps on turning.
It could be fixed by adding a timeout diamond. If the robot cant match the signals in 5000ms the wheels stop and it returns to main loop
Start steering
Measure signal strength from RIGHT and LEFT sensors
Which sensor has the strongest signal?
LEFT
RIGHT
Turn LEFT until signal strength differences are less than 5% of each other
Return to main loop
Turn RIGHT until signal strength differences are less than 5% of each other
Return to main loop
Strength differences <5% of each other
Keep all wheel speeds equal
Return to main loop